SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07
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COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

1. Compare Star Topology and Bus Topology:

Criteria	Star Topology	Bus Topology
Cost	More expensive	Less expensive
Cabling	Requires separate cables for each node	Uses a single backbone cable
Central Device	Requires a hub or switch	Does not require a central device
Reliability	More reliable	Less reliable
Performance	Better performance	Performance decreases as devices increase
Failure	Failure of one node does not affect others	Failure of backbone cable can affect the entire network

Star topology is generally more expensive than bus topology because every device is connected individually to a central hub or switch, requiring more cables. Bus topology is cheaper because all devices share a single backbone cable.

In terms of reliability, star topology is better because failure of one connected device does not affect the other devices. In bus topology, if the main backbone cable fails, the entire network can stop working. Star topology also provides better performance because communication is managed through the central device. In bus topology, performance decreases when more devices are added because they share the same communication medium.

Example: A computer laboratory may use star topology because each computer can be connected separately to a central switch.

Conclusion: Bus topology is suitable when low cost and simple installation are important, whereas star topology is preferable when better reliability and performance are required.

2. Compare Mesh Topology and Star Topology

Criteria	Mesh Topology	Star Topology
Fault Tolerance	Very high	Moderate
Reliability	Very high	High
Connections	Multiple connections between nodes	Each node connects to a central device
Failure Handling	Alternate paths are available	Individual node failure does not affect others
Central Device Failure	No single central device dependency	Hub/switch failure can stop the network
Suitability	Critical and highly reliable environments	General networks

Mesh topology provides very high fault tolerance because nodes are connected through multiple paths. If one link fails, data can use another available path, so communication can continue.

Star topology has moderate fault tolerance. If one individual node or cable fails, the remaining network can continue to operate. However, the central hub or switch is a critical point. If it fails, the entire network may stop functioning.

Therefore, mesh topology is more robust and reliable than star topology, particularly in environments where continuous communication is important.

Example: Mesh topology can be useful in critical environments where alternative communication paths are required.

Conclusion: Mesh topology provides greater reliability and fault tolerance, while star topology is simpler and depends on a central device.

3. Compare Bus Topology and Mesh Topology

Criteria	Bus Topology	Mesh Topology
Installation	Simple	Complex
Cabling	Requires a single backbone cable	Requires many connections
Cost	Low	High
Complexity	Low	High
Maintenance/Setup	Easier	More difficult
Scalability of Connections	Simple for small networks	Becomes complex as devices increase
Suitability	Small networks	Networks requiring high reliability

Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections. This makes it suitable for small networks where a simple and inexpensive setup is required.

Mesh topology is much more complex because every node must be connected to multiple other nodes. As the number of devices increases, the number of connections and cables also increases significantly. This makes mesh topology more time-consuming and costly to install.

Although mesh topology is more expensive and complex, its multiple connections provide high reliability and fault tolerance.

Example: A small network with only a few devices can use bus topology because it requires fewer cables and connections.

Conclusion: Bus topology is easier and cheaper to install, while mesh topology provides better reliability but requires greater installation effort and cost.

4. Compare Hybrid Topology and Star Topology

Criteria	Hybrid Topology	Star Topology
Scalability	Highly scalable	Moderately scalable
Expansion	Flexible	Limited by central device
Structure	Combines two or more topologies	Uses a central hub/switch
Hardware Requirement	Can vary depending on combined topologies	Depends mainly on hub/switch capacity
Flexibility	Very high	Moderate
Large Network Suitability	Very suitable	Suitable but may require additional hardware

Hybrid topology combines two or more different topologies, such as star, mesh, or bus. Because different topologies can be combined, the network can be expanded according to the organization's requirements.

Star topology is also scalable, but its expansion is limited by the number of ports available on the central hub or switch. When the available ports are used, additional hardware may be required to connect more devices.

Hybrid topology therefore provides greater flexibility and scalability compared with a basic star topology.

Example: An organization can combine star and bus topologies to create a hybrid network and expand different sections according to its requirements.

Conclusion: Hybrid topology is more flexible and scalable, while star topology provides simpler expansion within the capacity of its central device.

5. Compare Star, Mesh, Bus, and Hybrid Topologies Based on Maintenance

Criteria	Star	Mesh	Bus	Hybrid
Maintenance	Easy	Difficult	Moderate/Difficult	Moderate
Troubleshooting	Easy	Time-consuming	Difficult	Requires careful monitoring
Complexity	Moderate	Very high	Low	Moderate to high
Fault Identification	Easy	Complex	Difficult	Depends on combined topologies
Administration	Easier	More difficult	Difficult when backbone fails	Requires skilled administration

Star topology is relatively easy to maintain because problems can usually be identified and isolated to a particular node or cable. This makes troubleshooting easier.

Mesh topology is highly reliable but difficult to maintain because it contains a large number of connections. Finding and troubleshooting problems can therefore take more time.

Bus topology is simple in structure, but maintenance can become difficult when the backbone cable develops a fault. Since the backbone is shared by the network, locating a fault can be challenging.

Hybrid topology has moderate maintenance complexity. It provides flexibility by combining different topologies, but administrators need to monitor and manage the different network structures carefully.

Example: In a star network, if one computer stops communicating, the administrator can check that computer's cable or connection without necessarily affecting other computers.

Conclusion: Star topology is generally the easiest to maintain, while mesh topology is the most complex. Bus topology has simple construction but can be difficult to troubleshoot when the backbone fails. Hybrid topology provides flexibility but requires careful management.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion